

Methods for solving trigonometric equations

Factorise, substitute, or reduce to one function — three moves cover almost everything.

LESSON 82–85

4 × 40 MIN

ALGEBRA 10, §27

P1 5.6–5.7

LESSON CLOCK

20'

40'

40'

40'

35'

5

The three moves	20 min
Quadratic by substitution	40 min
Factorising	40 min
Homogeneous equations	40 min
Double angle and auxiliary angle	35 min
Homework	5 min

LEARNING OBJECTIVES

- Reduce an equation to a single trigonometric function.
- Solve quadratic equations in $\sin x$ or $\cos x$ by substitution.
- Solve homogeneous equations by dividing by $\cos x$.
- Use the identities for $\sin 2x$, $\cos 2x$ and $a \sin x + b \cos x$.

TEXTBOOK

National	pp. 271–288
Cambridge	Pure Mathematics 1, pp. 105–116
Workbook	Exercise 5E–5F

01

Explanation

The three moves

Every equation in this section is turned into $\sin x = a$ or one of its three relatives. There are only three ways to get there.

MOVE	RECOGNISE IT BY	EXAMPLE
substitute	one function, two powers	$2\sin^2x - 3\sin x + 1 = 0$
factorise	a common factor, or a product = 0	$\sin x \cos x - \sin x = 0$
reduce	two functions of the same angle	$2\cos^2x + \sin x = 2$

THE IDENTITY THAT DOES MOST OF THE WORK

$\sin^2x + \cos^2x = 1$

Read left to right it removes a $\cos^2$; read right to left it creates one. Almost every “reduce” step is this identity used in one direction or the other.

Quadratic by substitution

Example. $2\sin^2 x - 3\sin x + 1 = 0$. Put $t = \sin x$:

$$2t^2 - 3t + 1 = 0 \Rightarrow t = 1 \text{ or } t = \frac{1}{2}$$

Now solve each: $\sin x = 1 \Rightarrow x = \frac{\pi}{2} + 2\pi n$, and $\sin x = \frac{1}{2} \Rightarrow x = (-1)^n \frac{\pi}{6} + \pi n$.

CHECK EVERY ROOT AGAINST $|T| \leq 1$

$2\sin^2 x - \sin x - 6 = 0$ gives $t = 2$ or $t = -\frac{3}{2}$. Both are outside $[-1, 1]$, so the equation has **no** solutions. Do the check before solving.

When the equation contains $\sin^2 x$ and $\cos x$, replace first: $2\cos^2 x + \sin x = 2$ becomes $2(1 - \sin^2 x) + \sin x = 2$, that is $2\sin^2 x - \sin x = 0$ — now a quadratic in $\sin x$ that also factorises.

Factorising

When every term shares a factor, take it out — **never** divide by it.

$$\sin x \cos x = \sin x \Rightarrow \sin x (\cos x - 1) = 0$$

So $\sin x = 0$ or $\cos x = 1$, giving $x = \pi n$ together with $x = 2\pi n$ — and the second family is inside the first, so the answer is just $x = \pi n$.

DIVIDING BY $\sin x$ LOSES SOLUTIONS

Divide $\sin x \cos x = \sin x$ by $\sin x$ and you get $\cos x = 1$ alone — every solution with $\sin x = 0$ has vanished. Factor, set each factor to zero, keep both.

EQUATION	FACTORED	SOLUTIONS
$2\sin x \cos x = \cos x$	$\cos x(2\sin x - 1) = 0$	$\frac{\pi}{2} + \pi n; (-1)^n \frac{\pi}{6} + \pi n$
$\tan x \cdot \sin x = \tan x$	$\tan x(\sin x - 1) = 0$	$\pi n; \frac{\pi}{2} + 2\pi n$
$\sin 2x = \sin x$	$\sin x(2\cos x - 1) = 0$	$\pi n; \pm \frac{\pi}{3} + 2\pi n$

Homogeneous equations

An equation is **homogeneous of degree one** when every term has exactly one factor of $\sin x$ or $\cos x$:

$$a \sin x + b \cos x = 0$$

Here dividing is safe — by $\cos x$, because $\cos x = 0$ would force $\sin x = 0$ too, and the two are never zero together. The result:

$$a \tan x + b = 0 \Rightarrow \tan x = -\frac{b}{a}$$

Degree two works the same way. $3\sin^2 x - 4\sin x \cos x + \cos^2 x = 0$: divide through by $\cos^2 x$ to get $3\tan^2 x - 4\tan x + 1 = 0$, a quadratic in $\tan x$ with roots 1 and $\frac{1}{3}$.

HOMOGENEOUS, OR NOT?

Count the trigonometric factors in each term. $3\sin^2 x - 4\sin x \cos x + \cos^2 x$ — two, two, two: homogeneous. $3\sin^2 x - 4\sin x \cos x + 1$ — two, two, **zero**: not homogeneous, until you write $1 = \sin^2 x + \cos^2 x$ and it becomes so.

Double angle and the auxiliary angle

$$\sin 2x = 2 \sin x \cos x \quad \cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1$$

Three forms of $\cos 2x$, and the right one is the one that leaves a single function. $\cos 2x + \sin x = 0$ becomes $1 - 2\sin^2 x + \sin x = 0$, a quadratic in $\sin x$.

The auxiliary angle. An equation of the form $a \sin x + b \cos x = c$ is not homogeneous, because of the constant. Write the left side as a single sine:

$$a \sin x + b \cos x = R \sin(x + \alpha), R = \sqrt{a^2 + b^2}, \tan \alpha = \frac{b}{a}$$

Example. $\sqrt{3} \sin x + \cos x = 1$. Here $R = 2$ and $\alpha = \frac{\pi}{6}$, so $2 \sin(x + \frac{\pi}{6}) = 1$, that is

$$\sin(x + \frac{\pi}{6}) = \frac{1}{2}.$$

WHAT R TELLS YOU IMMEDIATELY

Since $|\sin| \leq 1$, the expression $a \sin x + b \cos x$ never leaves $[-R, R]$. So $\sqrt{3} \sin x + \cos x = 3$ has no solution — $R = 2 < 3$ — and the maximum of $3 \sin x + 4 \cos x$ is 5. One number answers a whole family of questions.

02 Worked examples

EXAMPLE 1 Solve $2\cos^2 x + 3 \sin x = 0$.

① $2(1 - \sin^2 x) + 3 \sin x = 0$

Reduce to one function.

② $2\sin^2 x - 3 \sin x - 2 = 0$

③ $t = 2$ (rejected) or $t = -\frac{1}{2}$

$|t| \leq 1$ kills the first.

④ $x = (-1)^{n+1} \frac{\pi}{6} + \pi n$

ANSWER

$$x = (-1)^{n+1} \frac{\pi}{6} + \pi n$$

EXAMPLE 2 Solve $\sin 2x = \cos x$.

① $2 \sin x \cos x - \cos x = 0$
Double angle first.

② $\cos x (2 \sin x - 1) = 0$
Factor — do not divide.

③ $\cos x = 0 \Rightarrow x = \frac{\pi}{2} + \pi n$

④ $\sin x = \frac{1}{2} \Rightarrow x = (-1)^n \frac{\pi}{6} + \pi n$

ANSWER

$$x = \frac{\pi}{2} + \pi n \text{ or } x = (-1)^n \frac{\pi}{6} + \pi n$$

EXAMPLE 3 Solve $\sin^2 x - 3 \sin x \cos x + 2 \cos^2 x = 0$.

① Homogeneous of degree 2; divide by $\cos^2 x$.

② $\tan^2 x - 3 \tan x + 2 = 0$

③ $\tan x = 1$ or $\tan x = 2$

④ $x = \frac{\pi}{4} + \pi n$ or $x = \arctan 2 + \pi n$

ANSWER

$$x = \frac{\pi}{4} + \pi n, x = \arctan 2 + \pi n$$

EXAMPLE 4 Solve $\sqrt{3} \sin x + \cos x = 1$ on $[0, 2\pi]$.

$$① \quad R = \sqrt{3+1} = 2, \tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$$

$$② \quad 2 \sin\left(x + \frac{\pi}{6}\right) = 1$$

$$③ \quad x + \frac{\pi}{6} = \frac{\pi}{6} \text{ or } \frac{5\pi}{6} \pmod{2\pi}$$

$$④ \quad x = 0 \text{ or } x = \frac{2\pi}{3}; \text{ also } x = 2\pi.$$

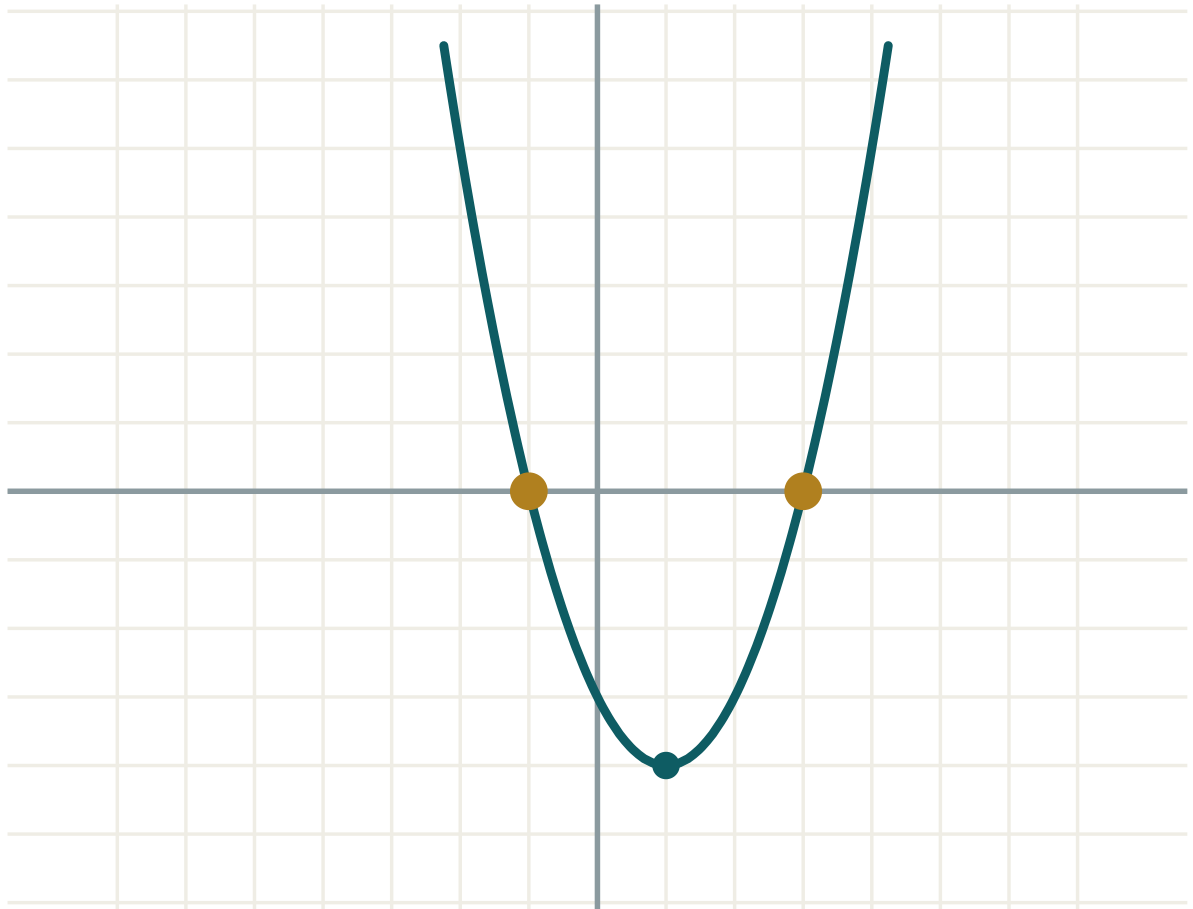
ANSWER

$$x = 0, \frac{2\pi}{3}, 2\pi$$

03 Interactive model

Put the three moves on the board as a checklist and refuse to start any question until the class has named the move.

The quadratic behind the substitution



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Substitution	O'rniga qo'yish	Подстановка
Homogeneous equation	Bir jinsli tenglama	Однородное уравнение
Factorisation	Ko'paytuvchilarga ajratish	Разложение на множители
Double angle	Ikkilangan burchak	Двойной угол
Identity	Ayniyat	Тождество
Auxiliary angle	Yordamchi burchak	Вспомогательный угол
Extraneous root	Chet ildiz	Посторонний корень
Domain restriction	Aniqlanish sohasi cheklovi	Ограничение области
Quadratic in $\sin x$	$\sin x$ ga nisbatan kvadrat	Квадратное относительно $\sin x$
Reduction	Keltirish	Приведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2\sin^2 x - \sin x - 6 = 0$ has:

two families

one family

no solution

four families

To solve $\sin x \cos x = \sin x$ you should:

divide by $\sin x$

factorise

square both sides

substitute $t = \cos x$

A homogeneous equation is solved by dividing by:

$\sin x$

the highest power of $\cos x$

2

nothing

The maximum of $3 \sin x + 4 \cos x$ is:

Which form of $\cos 2x$ suits $\cos 2x + \sin x = 0$?

$a \sin x + b \cos x = c$ has no solution when:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x (\cos x - 1) = 0$

ANSWER $x = \pi n$

2. Solve $\cos x (2 \sin x - 1) = 0$

ANSWER $\frac{\pi}{2} + \pi n; (-1)^n \frac{\pi}{6} + \pi n$

3. Substitute $t = \sin x$ in $2\sin^2 x - 3 \sin x + 1 = 0$

ANSWER $2t^2 - 3t + 1 = 0$

4. Roots of $2t^2 - 3t + 1 = 0$

ANSWER $t = 1, t = \frac{1}{2}$

5. Replace $\cos^2 x$ in $2\cos^2 x + \sin x = 2$

ANSWER $2\sin^2 x - \sin x = 0$

6. R for $3 \sin x + 4 \cos x$

ANSWER 5

7. R for $\sin x + \cos x$

ANSWER $\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Solve $2\sin^2 x - 3 \sin x + 1 = 0$

ANSWER $\frac{\pi}{2} + 2\pi n; (-1)^n \frac{\pi}{6} + \pi n$

2. Solve $2\cos^2 x + \sin x = 2$

ANSWER $\pi n; (-1)^n \frac{\pi}{6} + \pi n$

3. Solve $\sin 2x = \sin x$

ANSWER $\pi n; \pm \frac{\pi}{3} + 2\pi n$

4. Solve $\sin x + \cos x = 0$

ANSWER $x = -\frac{\pi}{4} + \pi n$

5. Solve $\sqrt{3} \sin x - \cos x = 0$

ANSWER $x = \frac{\pi}{6} + \pi n$

6. Solve $\cos 2x + \cos x = 0$ on $[0, 2\pi]$

ANSWER $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

7. Solve $\tan^2 x - 1 = 0$

ANSWER $x = \pm \frac{\pi}{4} + \pi n$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sin^2 x - 3 \sin x \cos x + 2 \cos^2 x = 0$

ANSWER $\frac{\pi}{4} + \pi n; \arctan 2 + \pi n$

2. Solve $\sqrt{3} \sin x + \cos x = 1$ on $[0, 2\pi]$

ANSWER $0, \frac{2\pi}{3}, 2\pi$

3. Solve $\sin x + \cos x = 1$ on $[0, 2\pi]$

ANSWER $0, \frac{\pi}{2}, 2\pi$

4. Solve $\cos 2x = 3 \cos x - 2$ on $[0, 2\pi]$

ANSWER $0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$

5. Solve $\sin 2x + \cos 2x = 1$ on $[0, 2\pi]$

ANSWER $0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}, 2\pi$

6. Find the maximum and minimum of $5 \sin x - 12 \cos x$

ANSWER 13 and -13

7. For which c has $\sin x + \sqrt{3} \cos x = c$ a solution?

ANSWER $|c| \leq 2$

07 Homework

Homework — 5 tasks

Name the move — substitute, factorise or reduce — before each solution.

1. Solve $2\sin^2 x + \sin x - 1 = 0$ in general and on $[0, 2\pi]$.
2. Solve $\sin 2x = \cos x$ and explain in one sentence why dividing by $\cos x$ would be wrong.
3. Solve $3\sin^2 x - 4 \sin x \cos x + \cos^2 x = 0$.

4. Write $5 \sin x + 12 \cos x$ in the form $R \sin(x + \alpha)$ and state its greatest value.
5. Solve $\cos 2x + 3 \sin x = 2$ on $[0^\circ, 360^\circ]$.